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Uncovering Shared Structures in Multiclass Classification

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Abstract

This paper suggests a method for multiclass learning with many classes by simultaneously learning shared characteristics common to the classes, and predictors for the classes in terms of these characteristics. We cast this as a convex optimization problem, using *trace-norm* regularization and study gradient-based optimization both for the linear case and the kernelized setting.

1. Introduction

In this paper we address the question of how to utilize hidden structure in order to improve multiclass classification accuracy. Our goal is to provide a mechanism for learning the underlying characteristics that are shared between the target classes. We demonstrate the benefit of extracting common characteristics within the powerful notion of large margin multiclass linear classifiers.

The challenge of accurate classification of an instance into one of a large number of target classes surfaces in many domains, such as object recognition, face identification, textual topic classification, and phoneme recognition. In many of these domains it is natural to assume that even though there are a large number of classes (e.g. different people in a face recognition task), classes are related and build on some underlying common characteristics. For example, many different mammals share characteristics such as a striped texture or an elongated snout, and people's faces can be identified based on underlying characteristics such as gender, being Caucasian, or having red hair. Recovering the true underlying characteristics of a domain can sig-

nificantly reduce the effective complexity of the multiclass problem, therefore transferring knowledge between related classes.

The obvious question that arises is how to select the feature mapping appropriate for a given task. One method to resolve this need is by manually designing a domain specific kernel. When the route of manual kernel design is not feasible one can attempt to learn a data specific feature mapping (Crammer et al., 2002). In practice, researchers often simply test several of the standard kernels in order to assess which attains better performance on a validation set. However, these approaches fail to provide a clear mechanism for utilizing existing underlying structures between the target classes. We would therefore like to find an efficient way to learn feature mappings that capture those underlying structures that characterize a given set of classes.

The observation that learning a hidden representation of some shared characteristics can facilitate learning has a long history in multiclass learning (e.g. Dekel et al. (2004)). This notion is often termed learning-to-learn or interclass transfer (Thrun, 1996). While some approaches assume some information on the shared characteristics is provided to the learner in advance (e.g. (Fink et al., 2006)), others rely on various heuristics in order to extract the shared features (e.g. (Torralba et al., 2004)).

Simultaneously learning the underlying structure between the classes and the class models is a challenging optimization task. Many of the heuristic approaches explored in the past aim at extracting powerful non-linear hidden characteristics. However, this goal often entails non-convex optimization tasks, prone to local minima problems. In contrast, we will focus on modeling the shared characteristics, as linear transformations of the input space. Thus, our model will postulate a linear mapping of shared features, followed by a multiclass linear classifier. We will show

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that such models can be efficiently learned in a convex optimization scheme, and albeit restricted to simple linear mappings, they can still significantly improve the accuracy of multiclass linear classifiers.

2. Formulation

The goal of multiclass classification is to learn a mapping $H : \mathcal{X} \rightarrow \mathcal{Y}$ from instances in $\mathcal{X}$ to labels in $\mathcal{Y} = \{1, \dots, k\}$. We consider linear classifiers over $\mathcal{X} = \mathbb{R}^n$, parametrized by a weight vector $W_y \in \mathbb{R}^n$ for each class $y \in \mathcal{Y}$, and which take the form:

$$H_W(x) = \operatorname{argmax}_{y \in \mathcal{Y}} W_y^t \cdot x \quad (1)$$

We wish to learn the weights from a set of m labeled training examples $(\mathbf{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, which we summarize in a matrix $X \in \mathbb{R}^{n \times m}$ whose columns are given by $\mathbf{x}_i$. Inspired by the large margin approach for classification, Crammer and Singer (2001) suggest learning the weights by minimizing a trade-off between an average empirical loss (to be discussed shortly) and a regularizer of the form:

$$\sum_y \|W_y\|^2 = \|W\|_F^2 \quad (2)$$

where $\|W\|_F$ is the Frobenius norm of the matrix W whose columns are the vectors W_y . The loss function suggested by Crammer *et al* is the maximal hinge loss over all comparisons between the correct class and an incorrect class:

$$\ell(W; (\mathbf{x}, y)) = \max_{y' \neq y} [1 + W_{y'}^t \cdot \mathbf{x} - W_y^t \cdot \mathbf{x}]_+ \quad (3)$$

where $[z]_+ = \max(0, z)$. For a trade-off parameter C , the weights are then given by the following learning rule:

$$\min_W \frac{1}{2} \|W\|_F^2 + C \sum_{i=1}^m \ell(W; (\mathbf{x}_i, y_i)) \quad (4)$$

For a binary classification problem, $\mathcal{Y} = \{1, 2\}$, this formulation reduces to the familiar Support Vector Machine (SVM) formulation (with $W_1 = -W_2 = \frac{1}{2} \mathbf{w}_{\text{svm}}$ at the optimum, and C appropriately scaled). For a larger number of classes, the formulation generalizes SVMs by requiring a margin between every pair of classes, and penalizing, for each training example, the amount by which the margin constraint it violated. Similarly to SVMs the optimization problem Eq. (4) is convex, and by introducing a “slack variable” for each example, it can be written as quadratic programming. It should be noted that while we choose to focus on the loss function of Crammer and Singer (2001), the methods we propose can be directly applied to other multiclass losses.

Recall that our goal is to attain a classifier W with improved generalization by extracting characteristics that are

shared among multiple classes. We restrict ourselves to modelling each common characteristic r as a linear function $F_r^t \mathbf{x}$ of the input vectors $\mathbf{x}$. The activation of each class y is then taken to be a linear function $G_y^t(F^t \mathbf{x})$ of the vector $F^t \mathbf{x}$ of common characteristics, instead of a linear function of the input vectors. Formally our model substitutes the weight matrix $W \in \mathbb{R}^{n \times k}$ with the product $W = FG$ of a weight matrix $F \in \mathbb{R}^{n \times p}$, whose columns define the p common characteristics, and $G \in \mathbb{R}^{p \times k}$, whose columns predict the classes based on the common characteristics:

$$H_{G,F}(x) = \operatorname{argmax}_{y \in \mathcal{Y}} G_y^t \cdot (F^t x) = \operatorname{argmax}_{y \in \mathcal{Y}} (FG)_y^t \cdot x \quad (5)$$

It should be emphasized that if F and G are not constrained in any way, the hypothesis space defined by Eq. (1) and by Eq. (5) is identical, since any linear transformations induced by applying F and then G , can always be attained by a single linear transformation W . We aim to show that nevertheless, regularizing the decomposition FG , as we discuss shortly, instead of the Frobenius norm of the weight matrix W , can yield a significant generalization advantage.

When the common characteristics F are known, we can replace the input instances $\mathbf{x}_i$ with the vectors $F^t \mathbf{x}_i$ and revert back to our original formulation Eq. (4), with the matrix G taking the role of the weight matrix. Each characteristic r is now a *feature* $(F^t \mathbf{x}_i)_r$ in this transformed problem. The challenge we address in this paper is of simultaneously learning the common characteristics (or latent features) F and the class weights G .

Increasing the norm $\|F_r\|$ allows smaller values of G_{yr} to yield the same prediction. Therefore, in order for the regularizer $\|G\|_F$ to be meaningful, we must also control the magnitude of F . We thus suggest to regularize, in addition to $\|G\|_F$, also $\sum_r \|F_r\|^2 = \|F\|_F^2$. This leads to the following learning rule:

$$\min_{F,G} \frac{1}{2} \|F\|_F^2 + \frac{1}{2} \|G\|_F^2 + C \sum_{i=1}^m \ell(FG; (\mathbf{x}_i, y_i)) \quad (6)$$

As we are accustomed to in large-margin methods, we do not have to limit the number of characteristics p . We can consider the rule Eq. (6) where the minimization is over matrices F, G of arbitrary inner dimensionality. We are relying here on the *norm* of F and G for regularization, rather than their *dimensionality*.

The optimization objective of Eq. (6) is non-convex, and involves matrices of unbounded dimensionality. However, instead of explicitly learning F, G , the optimization problem Eq. (6) can also be written directly as a convex learning rule for W . Following Srebro *et al.* (2005), we consider the

trace-norm of a matrix W :

$$\|W\|_\Sigma = \min_{FG=W} \frac{1}{2} (\|F\|_F^2 + \|G\|_F^2) \quad (7)$$

The trace-norm is a convex function of W , and can be characterized as the sum of its singular values (Boyd & Vandenberghe, 2004).

$$\|W\|_\Sigma = \sum_i |\gamma_i|, \quad (8)$$

Using Eq. (7), we can rewrite Eq. (6) as:

$$\min_W \|W\|_\Sigma + C \sum_{i=1}^m \ell(W; (\mathbf{x}_i, y_i)) \quad (9)$$

Furthermore, following Fazel et al. (2001) and Srebro et al. (2005), the optimization problem Eq. (9) can be formulated as a semi-definite program (SDP).

To summarize, we saw how learning to classify based on shared characteristics yields a learning rule in which the Frobenius-norm regularization is replaced with a trace-norm regularization.

3. Dualization and Kernelization

So far, we assumed we have direct access to the feature representation $\mathbf{x}$. However, much of the success of large-margin methods stems from the fact that one does not need access to the feature representation itself, but only to the inner product between feature vectors, specified by a *kernel function* $k(\mathbf{x}, \mathbf{x}')$. In order to obtain a kernelized form of trace-norm regularized multiclass learning, we first briefly describe the dual of Eq. (9), and how the optimum W can be obtained from the dual optimum.

By applying standard Lagrange duality we deduce the dual of Eq. (9) is given by the following optimization problem, which can also be written as a semi-definite program:

$$\begin{aligned} & \forall_{i,j \neq y_i} Q_{ij} \geq 0 \\ \max \sum_i (-Q_{iy_i}) \quad \text{s.t.} \quad & \forall_i (-Q_{iy_i}) = \sum_{j \neq y_i} Q_{ij} \leq c \\ & \|XQ\|_2 \leq 1 \end{aligned}$$

where $Q \in \mathbb{R}^{m \times k}$ denotes the dual Lagrange variable and $\|XQ\|_2$ is the spectral norm of XQ (i.e. the maximal singular value of this matrix). The spectral norm constraint can be equivalently specified as $\|(XQ)^t(XQ)\|_2 = \|Q^t(X^tX)Q\|_2 \leq 1$. This form is particularly interesting, since it allows us to write the dual in terms of the Gram matrix $K = X^tX$ instead of the feature representation X

explicitly:

$$\begin{aligned} & \forall_{i,j \neq y_i} Q_{ij} \geq 0 \\ \max \sum_i (-Q_{iy_i}) \quad \text{s.t.} \quad & \forall_i (-Q_{iy_i}) = \sum_{j \neq y_i} Q_{ij} \leq c \\ & \|Q^t K Q\|_2 \leq 1 \end{aligned} \quad (10)$$

Eq. (10) is a convex problem on Q that involves a semi-definite constraint (the spectral-norm constraint) on the matrix $Q^t K Q$ whose size is independent of the size of the training set, and only depends on the number of classes k (the size of Q and the number of quadratic interactions in $Q^t K Q$ do grow with the training set size, as in a standard SVM).

The following Representer Theorem describes the optimum weight matrix W in terms of the dual optimum Q , and allows the use of the kernel mechanism for prediction.

Theorem 1 *Let Q be the optimum of Eq. (10) and V be the matrix of eigenvectors of $Q^t K Q$, then for some diagonal $D \in \mathbb{R}^{k \times k}$, the matrix $W = X(QV^t D V)$ is an optimum of Eq. (9), with $\|W\|_\Sigma = \sum_r |D_{rr}|$.*

Proof Using complementary slackness and following arguments similar to those of Srebro et al. (2005), it can be shown that XQ and the optimum W of Eq. (9) share the same singular vectors. That is, if $XQ = USV$ is the singular value decomposition of XQ , then $W = UDV$ for some diagonal matrix D . Furthermore $D_{rr} = 0$ whenever $S_{rr} \neq 1$, i.e. $SD = D$. Note also that the right singular vectors V of $XQ = USV$ are precisely the eigenvectors of $(XQ)^t(XQ) = Q^t X^t X Q = Q^t K Q$. We can now express W as follows. First note that $W = UDV$. Since $D = SD$ we may express W as $USDV$. Since $VV^t = I$ we may further expand this expression to $USV V^t D V$. Finally, replacing USV with XQ we obtain $W = X(QV^t D V)$.

Corollary 1 *There exists $\alpha \in \mathbb{R}^{m \times k}$ s.t. $W = X\alpha$ is an optimum of Eq. (9)*

The situation is perhaps not as pleasing as for standard SVMs where the weight vector can be explicitly represented in terms of the dual optimum solution. Here, even after obtaining the dual optimum Q , we still need to recover the diagonal matrix D . However, substituting $W = XQV^t D V$ into Eq. (9), the first term becomes $\sum_r |D_{rr}|$, while the second is piecewise linear in $KQV^t D V$. We therefore obtain a linear program (LP) in the k unknown entries on the diagonal of D , which can be easily solved to recover D , and hence W . It is important to stress that the number of variables of this LP depends only on the number of classes, and not on the size of the data set, and that the entire procedure (solving Eq. (10), extracting V and recovering D) uses only the Gram matrix K and does not require

direct access to the explicit feature vectors X .

Even if the dual is not directly tackled, the representation of the optimum W guaranteed by Thm. 1 can be used to solve the primal Eq. (9) using the Gram matrix K instead of the feature vectors X , as we discuss in Section 5.

4. Learning a Latent Feature Representation

As alluded to above, learning F can be thought of as learning a latent feature space $F^t X$, which is useful for prediction. Since F is learned jointly over all classes, it effectively transfers knowledge between the classes. Low-norm decompositions were previously discussed in these terms by Srebro et al. (2005). More recently, Argyriou et al. (2007) studied a formulation equivalent to using the trace-norm explicitly for transfer learning between multiple tasks. Consider k binary classification tasks, and use W_j as a linear predictor for the j 'th task. Using an SVM to learn each class independently corresponds to the learning rule:

$$\min_W \sum_j \left(\frac{1}{2} \|W_j\|^2 + C \ell_j(W_j) \right) = \min_W \frac{1}{2} \|W\|_F^2 + C \sum_j \ell_j(W_j)$$

where $\ell_j(W_j)$ is the total (hinge) loss of W_j on the training examples for task j . Replacing the Frobenius norm with the trace norm:

$$\min_W \|W\|_\Sigma + C \sum_j \ell_j(W_j) \quad (11)$$

corresponds to learning a feature representation $\phi(\mathbf{x}) = F^t \mathbf{x}$ that allows good, low-norm prediction for all k tasks, where the linear predictor for task j , in this feature space, is given by G_j . After such a feature representation is learned, a new task can be learned directly using the feature vectors $F^t \mathbf{x}$ using standard SVM machinery, taking advantage of the transferred knowledge from the other, previously-learned, tasks.

In the multiclass setting, the predictors W_y are never independent, as even in the standard Frobenius norm formulation Eq. (4), the loss couples together the predictors for the different classes. However, the between-class transfer afforded by implicitly learning shared characteristics is much stronger. As will be demonstrated later, such transfer is particularly important if only a few number of examples are available from some class of interest.

Although this paper studies multiclass learning, the technical contributions, including the optimization approach, study of the dual problem, and kernelization, apply equally well also to the multi-task formulation Eq. (11).

It is interesting to note that we can learn a feature representation $\phi(\mathbf{x}) = F^t \mathbf{x}$ even when we are not given the feature

representation X explicitly, but only a kernel $\mathbf{k}$ from which we can obtain the Gram matrix $K = X^t X$. In this situation we do not have access to X , nor can we obtain F explicitly. As discussed above, what we *can* obtain is a matrix α such that $W = X\alpha$ is an optimum of Eq. (9). Let $W = UDV$ be the singular value decomposition of W (which we cannot calculate, since we do not have access to X). We have that $F = U\sqrt{D}$ is an optimum of Eq. (6). What we *can* calculate is the singular value decomposition of $\alpha^t K \alpha = \alpha^t X^t X \alpha = W^t W = V^t D^2 V$, and thus obtain D and V (but not U). Now, note that $D^{-1/2} V \alpha^t K = D^{-1/2} V (\alpha^t X^t) X = D^{-1/2} V W^t X = D^{-1/2} V V^t D U^t X = D^{1/2} U^t X = F^t X$, providing us with an explicit representation of the learned feature space that we can calculate from K and α alone.

In either case, we should note the optimum of Eq. (6) is not unique, and so also the learned feature space is not unique: if F, G is an optimum of Eq. (6), then $(FR), (R^t G)$ is also an optimum, for any unitary matrix $RR^t = I$. Instead of learning the explicit feature representation $\phi(\mathbf{x}) = F^t \mathbf{x}$, we can therefore think of trace-norm regularization as learning the implied kernel $\mathbf{k}_\phi(\mathbf{x}', \mathbf{x}) = \langle F^t \mathbf{x}', F^t \mathbf{x} \rangle$. Even when F is rotated (and reflected) by R , the learned kernel $\mathbf{k}_\phi$ is unaffected.

5. Optimization

The optimization problem Eq. (9) can be formulated as a semi-definite program (SDP) and off-the-shelf SDP solvers can be used to recover the optimal W . However, such solvers based on interior point methods scale poorly with the size of the problem and typically cannot handle problems with more than several hundred dimensions, classes and training points (10^4 variables). Hence, we choose to optimize Eq. (9) using simple, but powerful gradient-based methods.

5.1. Gradient Based Optimization

The optimization problem Eq. (9) is non-differentiable and so not immediately amenable to gradient-based optimization. In order to perform the optimization, we consider a smoothed approximation to Eq. (9).

We begin by replacing the trace-norm with a smooth proxy. Eq. (8) characterizes the trace-norm as the sum of the singular values of W . Although the singular values are non-negative, the absolute value in Eq. (8) emphasizes the reason the trace-norm is non-differentiable. In order to obtain a smooth approximation to the trace-norm, we replace the non-smooth absolute value with a smooth function g defined as,

$$g(\gamma) = \begin{cases} \frac{\gamma^2}{2r} + \frac{r}{2} & \gamma \leq r \\ |\gamma| & \text{otherwise} \end{cases}.$$

Where r is some predefined cutoff point. Fig. 1 illustrates the function g and the effect of the parameter r . We can easily see that g is continuously differentiable, and that $\forall x : |g(x) - |x|| \leq \frac{r}{2}$. Our smoothed proxy for the trace norm thus replaces Eq. (8) with

$$\|W\|_S = \sum_i g(\gamma_i) , \quad (12)$$

where γ_i are the singular values of W . Using the chain rule, we can calculate its gradient as,

$$\frac{\partial \|W\|_S}{\partial W} = U g'(D) V \quad (13)$$

where $W = UDV$ is the SVD of W and $g'(D)$ is an element-wise computation of the derivative g' of g on the diagonal of D .

We now turn our attention on the non-differentiable multiclass hinge-loss of Eq. (3). Since neither the hinge $\llbracket \cdot \rrbracket_+$ nor the $\max$ operators are differentiable we employ an adaptation of the log-loss for the multiclass setting, with a parameter γ controlling its sharpness (Zhang & Oles, 2001; Dekel et al., 2003),

$$\ell_S(W; (\mathbf{x}_i, y_i)) = \frac{1}{\lambda} \log \left(1 + \sum_{r \neq y_i} e^{\lambda \cdot (1 + W_{r \cdot} \mathbf{x}_i - W_{y_i \cdot} \mathbf{x}_i)} \right).$$

This is a convex and continuously differentiable function of W which approaches the multiclass hinge-loss as $\lambda \rightarrow \infty$ (Fig. 1). In summary, instead of Eq. (9) we consider the following optimization problem:

$$\min_W \|W\|_S + C \sum_{i=1}^m \ell_S(W; (\mathbf{x}_i, y_i)) \quad (14)$$

which is a convex and continuously differentiable function.

Fig. 2-left shows how optimization of the smoothed objective Eq. (14) approximately optimizes Eq. (9). We generated 160 training instances with 16 classes and 16-dimensional feature vectors using a random 16×16 weight matrix. For each value of γ , and a fixed $r = 0.01$ we compared the weight matrix W recovered using conjugate gradient descent on Eq. (14) to the optimizer of Eq. (9) found using an interior point SDP solver (we used SDP3 which outperformed other solvers such as SeDuMi and SDPAM). The figure plots the value of the original (non-smooth) objective of both solutions. For large values of γ , the smoothed optimization solves the original problem with very good accuracy.

Fig. 2-right describes the gained performance using the gradient based smooth objective Eq. (14) while gradually increasing the number of instances from 80 to 1000. It is apparent that even for relatively small number of instances,

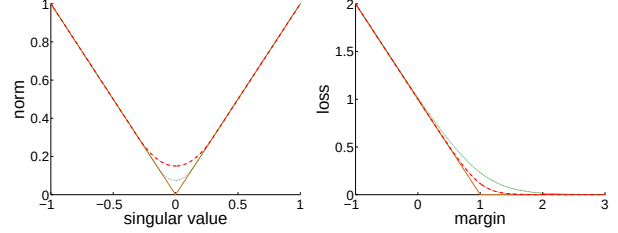


Figure 1. Left: The smoothed absolute value function g . Smaller values of r translate to a sharper function and a better estimate of the absolute values. Right: The binary version of the log-loss in comparison with the binary hinge-loss. Larger values of λ increase the accuracy of the log-loss approximation.

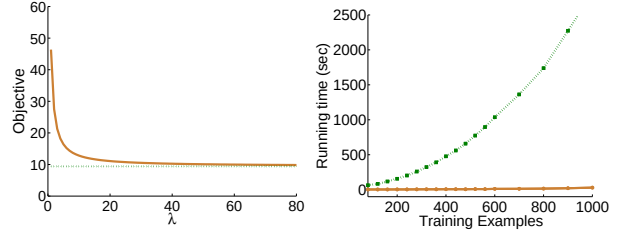


Figure 2. Left: The values of the original (non-smooth) optimization objective Eq. (9) for minima of the smoothed objective Eq. (14) as a function of the smoothing parameter λ (solid) compared to the true optimum of Eq. (9) (dotted). Right: Running times of SDP solver for Eq. (9) (dotted) vs the gradient based method for solving Eq. (14) (with $\lambda = 20$) as a function of the number of training instances.

the SDP optimization becomes unreasonably slow. In contrast, the gradient based optimization easily scales to fairly large training sets.

5.2. Kernelized Gradient Optimization

We now turn to devising a gradient-based optimization approach appropriate when only the Gram matrix $K = X^t X$ is available, but not the feature vectors X themselves. Corollary 1 assures us that the optimum of Eq. (9) is of the form $X\alpha$, and so we can substitute $W = X\alpha$ into Eq. (14) and minimize over α . To do so using gradient methods, we need to be able to compute both the smoothed objective and its derivative from K and α alone, without reference to X explicitly.

We first tackle the smoothed trace norm of $X\alpha$: Let $X\alpha = UDV$ denote the SVD of $X\alpha$ then the SVD of $\alpha^t K \alpha$ is given by $V^t D^2 V$. We can thus recover D from the SVD of $\alpha^t K \alpha$, and use Eq. (12) to calculate $\|X\alpha\|_S$.

In order to compute the gradient of $\|X\alpha\|_S$ with respect to α , we calculate:

$$\frac{\partial \|X\alpha\|_S}{\partial \alpha} = X^t \frac{\partial \|X\alpha\|_S}{\partial X\alpha} = X^t U g'(D) V$$

inserting $D(VV^t)D^{-1} = DID^{-1} = I$:

$$\begin{aligned} &= X^t U (D V V^t D^{-1}) g'(D) V \\ &= X^t (U D V) V^t D^{-1} g'(D) V \end{aligned}$$

and since $X\alpha = UDV$:

$$= X^t (X\alpha) V^t D^{-1} g'(D) V = K\alpha V^t D^{-1} g'(D) V \quad (15)$$

Recall that both V and D can be obtained from the SVD of $\alpha^t K \alpha$, and so Eq. (15) provides a calculation of the gradient in terms of K and α . Thus, we can efficiently apply our gradient based optimizations to a kernel $\mathbf{k}(\mathbf{x}, \mathbf{x}')$.

6. Spectral Properties of Trace Norm Regularization

One way to appreciate the difference between the Frobenius norm and the trace norm of a matrix W is by observing that the squared Frobenius norm equals the sum of the squared singular values, $\sum_i \gamma_i^2$, while the trace norm is the sum of the singular values themselves, $\sum_i \gamma_i$. Thus, choosing to minimize $\|F\|_F^2 + \|G\|_F^2$ rather than $\|W\|_F^2$, imposes a regularization preference for an L_1 norm on the spectrum of W (rather than an L_2 norm). When the various target classes share common characteristics we expect the spectrum of W to be non-uniform, since a large portion of the spectrum must be concentrated on few eigenvalues. In these cases the L_2 spectrum regularization imposed by the Frobenius norm will tend to attenuate the spectrum. In contrast, the L_1 spectrum regularization imposed by the trace norm does not share this tendency, and is thus better suited to preserve underlying structures of characteristics that are shared between the target classes.

In order to illustrate this effect we generated 100 classes over R^{120} and randomly sampled 4500 training instances from a 120-dimensional normal distribution. A 120×100 matrix W^* was then used to label the data, by choosing for each instance $\mathbf{x}$ the label $y = \operatorname{argmax}_{r \in \mathcal{Y}} W_r^* \cdot \mathbf{x}$. The matrix W^* was selected to have a sigmoidal pattern of singular values, depicted in the dashed spectrum on Fig. 3. We then recovered two matrices W_F and W_Σ using the Frobenius norm optimization from Eq. (4) and the trace norm optimization from Eq. (9). The generalization error over 500 new test instances, was significantly higher for W_F (47%) than for W_Σ (31%). The spectrum of the two learned models is depicted in Fig. 3. It could be observed that Frobenius based regularization leads to the attenuated spectrum of W_F .

A question may arise whether it was possible to encourage the underlying common structure between the classes by applying a dimensionality reduction procedure to the

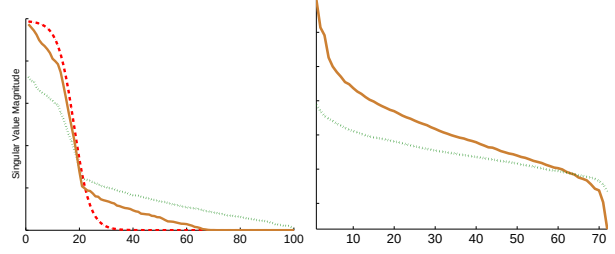


Figure 3. Spectra of learned matrices in the synthetic (left) and real (right) experiments. The weight matrix resulting from trace regularization (solid), and the weight matrix resulting from Frobenius regularization (dotted). The weight matrix that generated the data W^* (dashed) in the synthetic experiment only.

weight matrix. In order to show this is not necessarily the case, we repeated the experiment described above, but W^* was selected to have the singular values form a harmonic series $(\frac{1}{1}, \frac{1}{2}, \dots, \frac{1}{100})$. We similarly recovered two matrices W_F and W_Σ using the Frobenius norm optimization and the trace norm optimization. It was observed that the generalization error over 500 new test instances, was significantly higher for W_F (26%) than for W_Σ (17%).

Next, a singular value decomposition was performed on W_Σ and W_F followed by reconstructing these matrices using the p leading singular values and vectors ($p = 1, 2, \dots, 100$). Performance of the reconstructed weight matrices was evaluated on the test set. It was observed that any SVD dimensionality reduction deteriorated the test performance. Moreover, the generalization error for the reduced W_F was consistently worse than the performance of the reduced W_Σ . It could therefore be concluded that post-hoc dimensionality reduction could not attenuate the importance of finding the underlying structure as an integral part of the learning procedure.

7. Experiments

7.1. Experiment I: Letter Recognition

By analyzing over 100 writing systems, Changizi and Shimojo (2005) have demonstrated the fact that each writing system can be characterized by a set of underlying strokes. Therefore, our first experiment focuses on recognition of the 26 characters made available in the UCI *letter* dataset. The data was composed of 2000 instances, roughly distributed over the 26 classes. The data was partitioned to three sets: 1000 were used as a training set, 500 were held out and used to select the optimal value of C and 500 were used as a test set. Data was represented using a Gaussian kernel with $\sigma = 0.07$.

We then recovered two matrices W_F (Frobenius norm regularization) from Eq. (4) and W_Σ (trace norm regularization) from Eq. (9). The trade-off parameter C was deter-

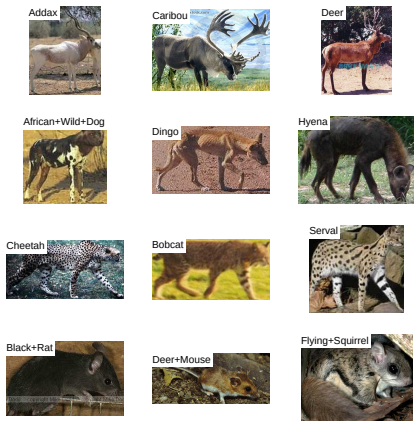


Figure 4. Representative images of Deer (Addax, Caribou, Common Deer), Canines (African Wild Dog, Dingo, Hyena), Felines (Cheetah, Bobcat, Serval), and Rodents (Black Rat, Deer Mouse, Flying Squirrel).

mined exhaustively by searching over 15 values between 2^{-9} and 2^5 . The value was later fine tuned by searching within a smaller window within $C \cdot 2^{-1.5}$ and $C \cdot 2^{1.5}$. All values were tested on the fixed holdout set. Performance was evaluated over 500 new test instances, and the generalization error was significantly higher for W_F (10.1%) than for W_Σ (8.7%).

7.2. Experiment II: Mammal Recognition Dataset

Our second experiment focused on the challenging task of classifying mammal images. We chose the 72 mammals that have at least 12 profile instances in the mammal benchmark made available by Fink and Ullman (2007). Of these, approximately 1,000 images were used for training and a similar number were used for testing. The test set was further partitioned, where half was held out and used to select C and the rest were used for testing. The number of instances of each class varied significantly from 6 to 30 training examples. It should be noted that the 72 target classes are expected to share many common characteristics due to genetic resemblance and evolutionary convergence. Four genetically related families (Deer, Canines, Felines and Rodents), are depicted in Fig. 4.

We build upon the comparison performed in Zhang et al. (2006) in selecting an image representation suitable for the high degree of intraclass variability present in the mammal dataset. This representation is based on extracting a visual signature from the images. The visual signatures include 40 clusters of local descriptors, extracted from interest regions of the image. The resulting signatures are compared using an Earth Moving Distance (EMD) Kernel. The EMD distance between signature-A and signature-B is found by solving the transportation problem, namely, by finding the

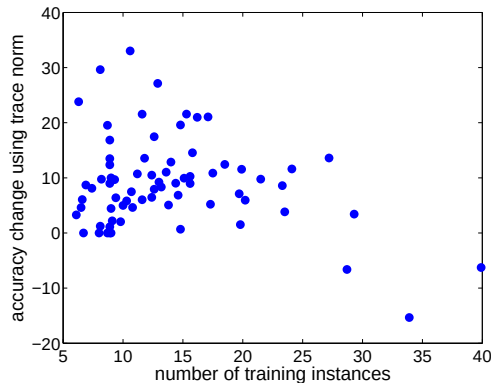


Figure 5. The gain in performance entailed by choosing trace norm regularization over Frobenius norm regularization, as a function of the number of training instances available in each mammal class.

minimal Euclidean distance necessary for converting the descriptors in signature-A to be identical to the descriptors of signature-B (for details see Zhang et al. (2006) and the references within).

Using the above representation we learned the two matrices W_F (Frobenius norm regularization) and W_Σ (trace norm regularization). The trade-off parameter C was determined using the same procedure used in Experiment 7.1. The accuracy of the multiclass SVM based on trace norm regularization (33%) is observed to be higher than that attained using the Frobenius norm regularization (29%).

In the previous sections it was suggested that learning F can be thought of as learning a latent feature space $F^t X$, which is useful for prediction. Since F is learned jointly over all classes, it can be thought of as transferring knowledge between the classes. Under these conditions a new class can be acquired from fairly few training examples. We therefore predict that classes with few training examples will, on average, gain more from applying trace norm regularization. This effect is depicted in Fig. 5. Specifically, it could be observed that of the few classes that gain from Frobenius regularization, four are of the top six most frequent mammals.

In order to verify this phenomenon we selected one of the most frequent classes (Wombats), which contains 30 training examples and repeatedly relearned W_F and W_Σ while reducing the number of wombat examples to 24, 18, and 12. Under these conditions the accuracy of correct classification of wombats naturally deteriorated, but the effect was noticeably less severe for the trace norm regularization. While the Frobenius norm regularization performed better when all 30 instances were available during learning (by 2.2%), when 24 instances were available the gap had narrowed to 1.2%. When even fewer examples were available the leads were reversed and the trace norm outperformed

the Frobenius norm by 1.4% for 18 instances and 3.7% for 12 instances. It should be noted that the false alarm rate over the remaining classes remained fairly constant. These results suggest that the learned common characteristics can indeed facilitate the acquisition of a novel class when only few examples are available for training.

Finally, the spectrum of the two learned models (Fig. 3), depicts the fact that Frobenius based regularization leads to the attenuated spectrum of W_F . It might be suggested that this effect manifests the advantage of trace norm regularization in preserving underlying structure between the mammal classes.

8. Discussion

We studied a learning rule for multiclass learning in which the magnitude of the factorization of the weight matrix is regularized, rather than the magnitude of the weights themselves. This is equivalent to regularizing the trace-norm of the weight matrix, instead of its Frobenius norm. We showed how this formulation can be kernelized, and solved efficiently either with direct access to the feature vectors or in a kernelized setting. We demonstrated the effectiveness of the formulation, particularly for classes with only a few available training examples.

The multiclass formulation we study is a special case of a general family of trace-norm regularized learning rules, where some general loss associated with the activation matrix $W^t X$ replaces our multiclass loss,

$$\min_W \|W\|_\Sigma + C \cdot \text{loss}(W^t X). \quad (16)$$

Maximum Margin Matrix Factorization (Srebro et al., 2005) can be seen as a degenerate case of Eq. (16) where $X = I$ and the loss function decomposes over the entries of W . More recently, Argyriou et al. (2007) studied a multi-task learning rule which can be shown to be equivalent to Eq. (16) (again with a decomposable loss function, as appropriate for the multi-task setting). Argyriou et al reach a different, but equivalent, formulation of the problem, relying on explicit access to the feature vectors X , and suggest an optimization approach which requires iteratively solving multiple SVM problems. We believe the formulation Eq. (16) is more direct and lends itself better to gradient-base optimization, which can be applied also for the multi-task setting. Our results on dualization, kernelization and representation of the learned latent feature space apply also to the multi-task setting studied by Argyriou et al, as well as to the general family of Eq. (16).

Another related learning rule using trace-norm regularization was studied by Abernethy et al. (2006). In their work, feature vectors are available as both “column” features (the matrix X) and “row” features (the matrix Z). The predic-

tion matrix is thus $ZW^t X$, rather than $W^t X$ in Eq. (16). However, the trace-norm regularization is applied to the prediction matrix $ZW^t X$, rather than to the weight matrix.

In this paper we suggested an efficient method to extract the underlying structures that characterize a set of target classes. We believe that this approach is part of a trend that emphasizes the importance of sharing representational knowledge in order to enable large scale classification.

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